Roadmap for climate-neutral ships - Emissions pathway

A.P.Møller - Maersk's
World's first carbon neutral liner in operation (Bio-methanol / VLSFO)

EU-funded ShipFC project
World's ammonia-powered fuel cell on offshore vessel with 2MW (Viking Energy, Equinor)

ESVAGT - Ørsted
World's first zero emission service operation vessel (SOV) operates on green fuels (e-methanol)

AMOGY - World's First Ammonia-Powered, Zero-Emission Ship
World's first zero emission ammonia-powered, zero-emission tugboat

Wallenius Wilhelmsen
World's first wind-powered RoRo (Orcelle Wind)

HyShip project
New roll-on/roll-off running on liquid green hydrogen (Topeka, Enova funding)

Ulstein – Nedstack - GE
Zero-Emission DP2 Construction Support Vessel Full LH_2-powered fuel cell with 2MW (Ulstein SX190)

A.P.Møller-Maersk's
Zero CO_2 emissions per container transported (today 46%)

Business-as-usual
(Based on predictions by IMO due to increasing sea traffic, 4% annual growth)

Technical improvements

Operational improvements

Wind propulsion

Battery

Other technologies / fuels

Hydrogen + Ammonia

Methanol

LNG / LPG

GHG reduction by

Alternative Technologies / fuels for replacing conventional fuels

Emissions pathway

- **Energy efficiency technology** (wind assistance, air lubrication, waste heat recovery, …)
- **Energy saving device** (wake equalizing duct, boss cap fin, …)
- **GHG-Emissions reduction devices** (Scrubbers, …)

- **Operational and environmental performance of the vessel** (fuel comsuption, speed reduction, transported cargo and distance travelled)

- **Plasma technology produces a synthetic natural gas (SNG)**
- **Anaerobic digestion of varying biomass sources to produce biomethane**
- **Carbon capture and storage on-board**

2008

2020

2022

2025

2030

2050

LOHC-testing offshore
200kW unit on-board Edda Ferd (ENOVA funding)

EU ETS 2024 to 2026 *
CO_2 tax for maritime
40% → 70% → 100%
Including CH_4 & N_2O

IMO GHG
CO_2 Reduction 40% compared to 2008
EU 2030 target
40% → 55%
Reduction of CO_2 compared 1990
Norway 2030
CO_2 tax (x 2) for most industries

LOHC-Ready on-Board (SHIP-AH2OY project)
Østensjø 1st CSOV with Hybrid power propulsion system

MPEC 80 Revision
mid-2023
100% decarbonisation by 2050?

IMO GHG (2050)
GHG Reduction 50% compared to 2008
CO_2 Reduction 70% compared to 2008
EU 2050 target
Net zero

IMO's GHG strategy

Zero emissions
As soon as possible

International Maritime Organization (IMO)

Scenario-based assessment of multiple source::
- Maritime forecast to 2050, Energy Transition Outlook 2020, DNV GL
- Assessment of selected alternative fuels and technologies , DNV GL June 2019
- Low carbon shipping towards 2050, DNV GL
- The maritime energy transition: Power-to-X as a key element for green shipping, MTU-Webinar, 12.11.2020
- Roadmap for Zero-Emission of International Shipping, MLIT Japan

Emissions

Time

Short term measures by IMO
EEDI, EEXI, **SEEMP (CII, AER, EEOI)**

Mid- and long term measures by IMO
Methane emission, **EEDI (Phase 4)** / Alternative fuels

*Designed by Juan M. Carvajal Ramos

on-board